10x data Annotation

Abinet Joan

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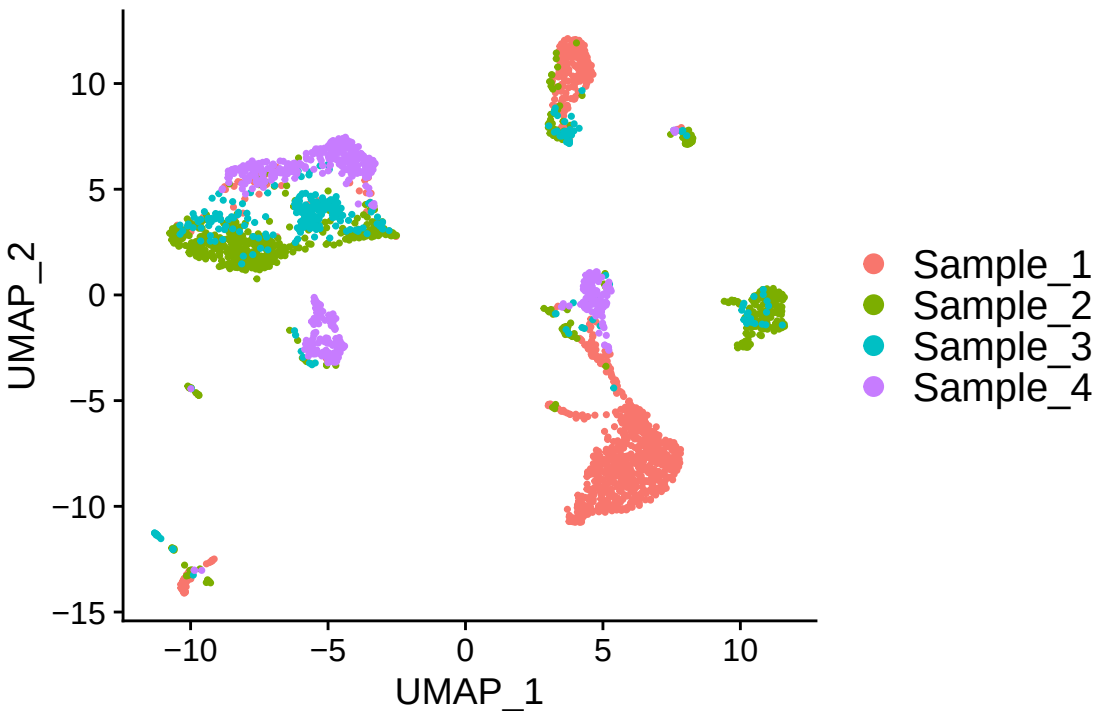
Loading packages

```
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(patchwork))
suppressMessages(library(ggplot2))
suppressMessages(library(RColorBrewer))
suppressMessages(library(dittoSeq))
suppressMessages(library(ComplexHeatmap))
suppressMessages(library(formatR))
```

Loading data

```
Cell_10xAnnotate <- readRDS("../10X-1-Processing/lavage_cells10x.rds")
```

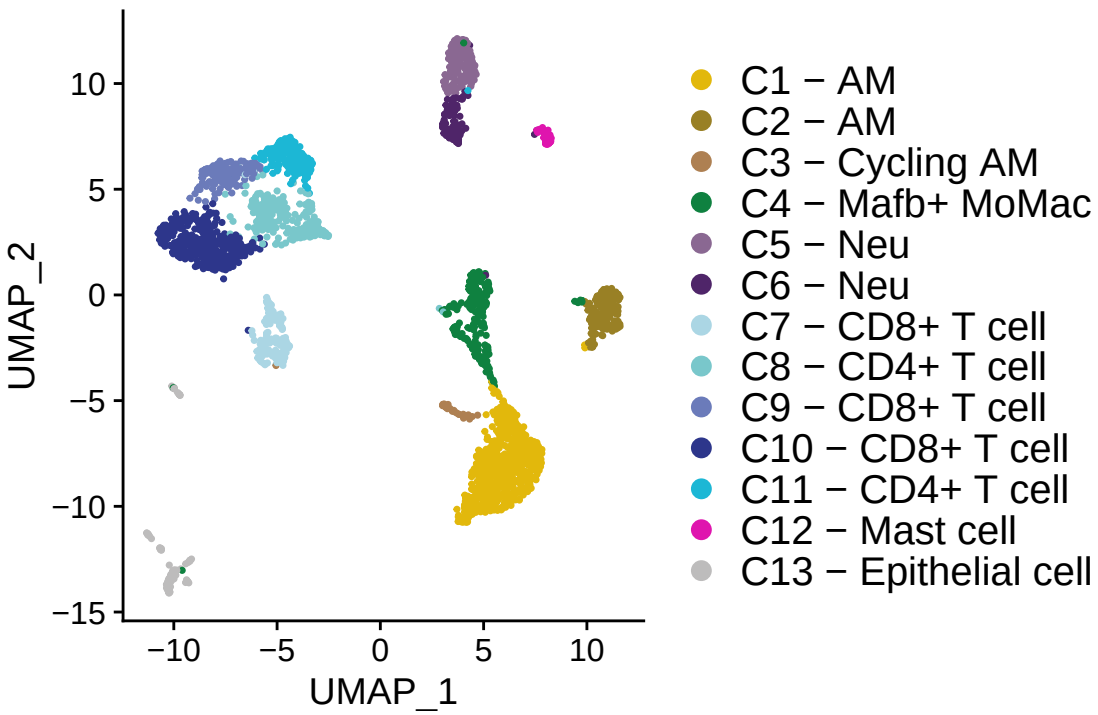
```
DimPlot(Cell_10xAnnotate, reduction = "umap", group.by = "orig.ident")+ theme(legend.text
= element_text(size = 15)) + labs(title = NULL)
```




```
#ggsave("Umap_PerSample.pdf", width = 8, height = 4)
```

Annotation

```
Cell_10xAnnotate$Annotation <- Cell_10xAnnotate$seurat_clusters
levels(Cell_10xAnnotate$Annotation) <- c("C1 - AM", "C10 - CD8+ T cell", "C8 - CD4+ T
cell", "C4 - Mafb+ MoMac", "C11 - CD4+ T cell", "C5 - Neu", "C2 - AM", "C7 - CD8+ T
cell", "C9 - CD8+ T cell", "C13 - Epithelial cell", "C6 - Neu", "C12 - Mast cell", "C3 -
Cycling AM")

Cell_10xAnnotate$Annotation <- factor(Cell_10xAnnotate$Annotation, levels =  c("C1 - AM",
"C2 - AM", "C3 - Cycling AM", "C4 - Mafb+ MoMac", "C5 - Neu", "C6 - Neu", "C7 - CD8+ T
cell", "C8 - CD4+ T cell", "C9 - CD8+ T cell", "C10 - CD8+ T cell", "C11 - CD4+ T cell",
"C12 - Mast cell",  "C13 - Epithelial cell"))


# Je modifie les couleurs pour qu'elles soient également à celle du HIVE
col <- c("#E2B80C", "#998025", "#AE8052", "#0F8140",  "#8A6892", "#4F2569", "#ABD6E4",
"#79C7CB", "#6B7BBA", "#2D368B", "#1CB7D5", "#DF15AE", "#BDBCBC")

DimPlot(Cell_10xAnnotate, group.by = "Annotation", cols = col)+ theme(legend.text    =
element_text(size = 15)) + labs(title = NULL)
```



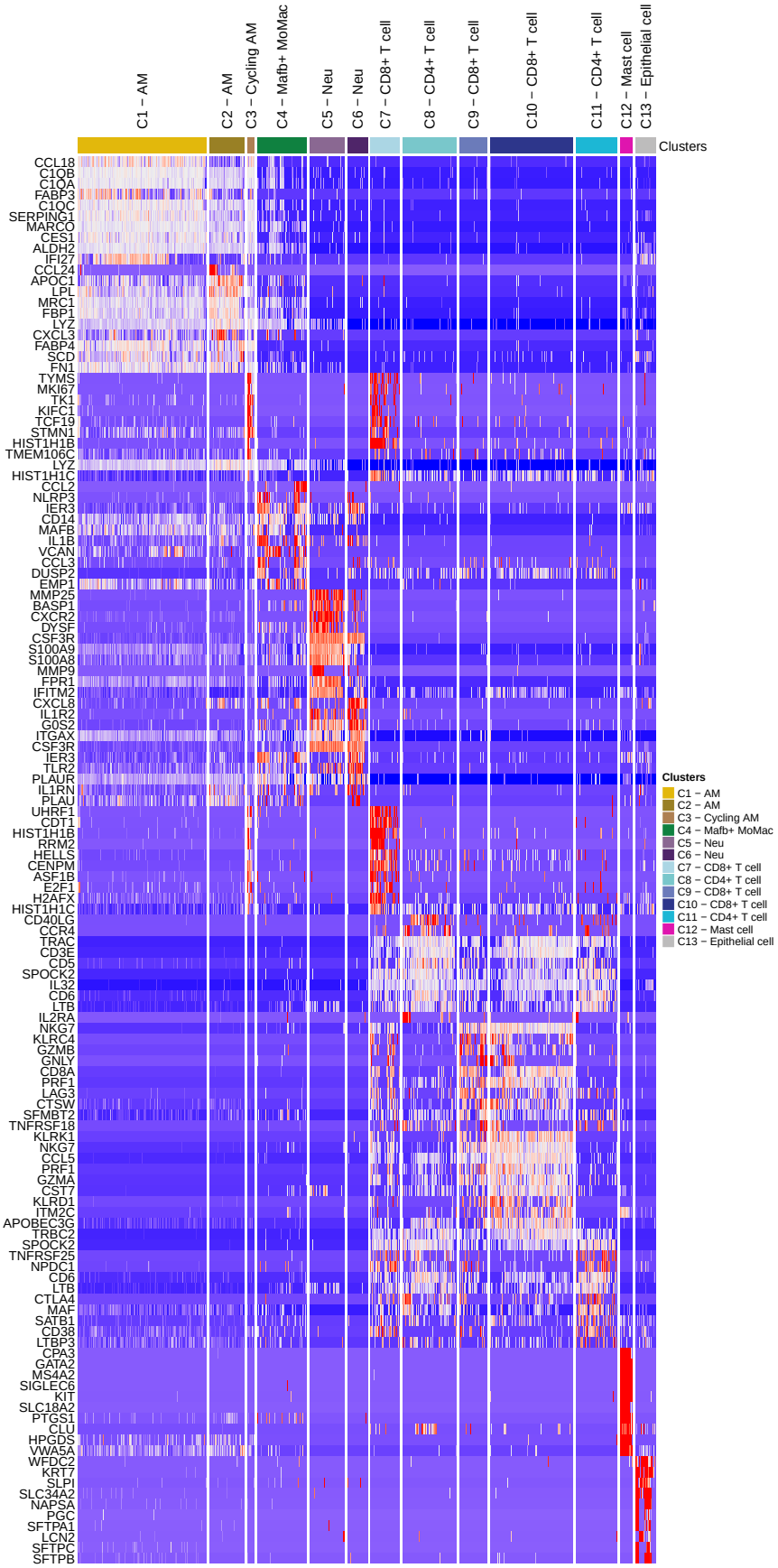

```
#ggsave("Umap_Annotation.pdf", width = 8, height = 4)
```

```
Idents(Cell_10xAnnotate) <- "Annotation"

lavage.markers <- FindAllMarkers(Cell_10xAnnotate, only.pos = TRUE, min.pct = 0.25)

lavage.markers %>%
    group_by(cluster) %>%
    top_n(n = 10, wt = avg_log2FC) -> top10

mat <- as.matrix( GetAssayData(object = Cell_10xAnnotate, slot =
"data")[as.character(top10$gene),])

df <- as.data.frame(Cell_10xAnnotate$Annotation)
colnames(df) <- "Clusters"
color_df <- list(Clusters = c("C1 - AM" = "#E2B80C",
                              "C2 - AM" = "#998025",
                              "C3 - Cycling AM" = "#AE8052",
                              "C4 - Mafb+ MoMac" = "#0F8140",
                              "C5 - Neu" = "#8A6892",
                              "C6 - Neu" = "#4F2569",
                              "C7 - CD8+ T cell" = "#ABD6E4",
                              "C8 - CD4+ T cell" = "#79C7CB",
                              "C9 - CD8+ T cell" = "#6B7BBA",
                              "C10 - CD8+ T cell" = "#2D368B",
                              "C11 - CD4+ T cell" = "#1CB7D5",
                              "C12 - Mast cell" = "#DF15AE",
                              "C13 - Epithelial cell" = "#BDBCBC"))


Heatmap<- Heatmap(t(scale(t(mat))), show_column_names = F,
```

```
        column_split = Cell_10xAnnotate$Annotation,
        cluster_column_slices = F,
        cluster_rows = F,
        top_annotation = HeatmapAnnotation(df = df, col = color_df),
        use_raster = F,
        show_heatmap_legend = F,
        show_column_dend = F,
        column_title_rot = 90,
        row_names_side = "left")

# tidyHeatmap::save_pdf(Heatmap, "Heatmap_10x.pdf", width = 35, height = 50, units ="cm")
```

Heatmap top genes per clusters

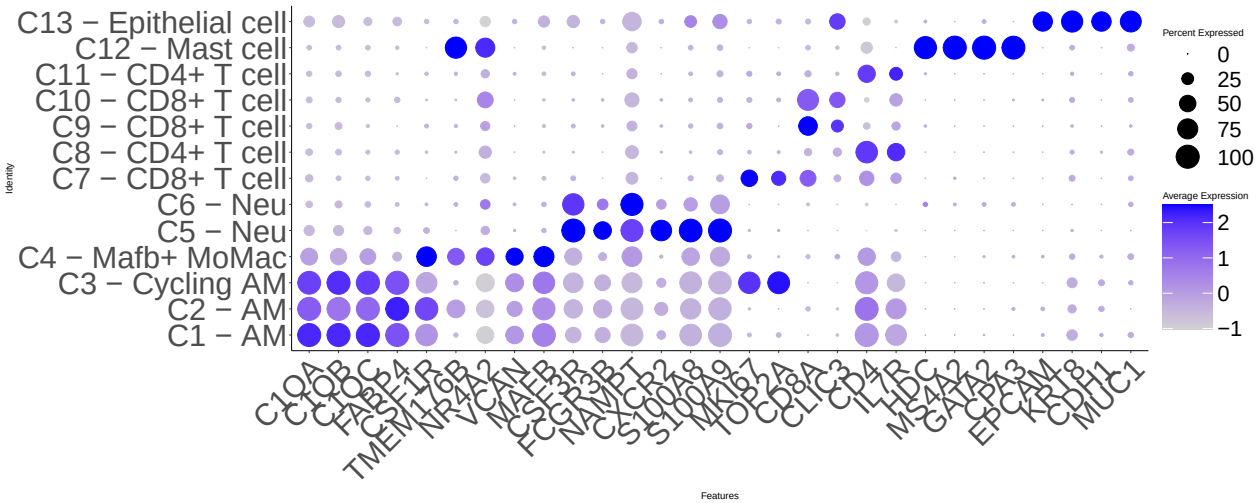
```
Heatmap
```


C1 – AM
C2 – AM
C3 – Cycling AM
C4 – Mafb+ MoMac
C5 – Neu
C6 – Neu
C7 – CD8+ T cell
C8 – CD4+ T cell
C9 – CD8+ T cell
C10 – CD8+ T cell
C11 – CD4+ T cell
C12 – Mast cell
C13 – Epithelial cell
Clusters
CCL18
C1QB
C1QA
FABP3
C1QC
SERPING1
MARCO
CES1
ALDH2
IFI27
CCL24
APOC1
LPL
MRC1
FBP1
LYZ
CXCL3
FABP4
SCD
FN1
TYMS
MKI67
TK1
KIFC1
TCF19
STMN1
HIST1H1B
TMEM106C
LYZ
HIST1H1C
CCL2
NLRP3
IER3
CD14
MAFB
IL1B
VCAN
CCL3
DUSP2
EMP1
MMP25
BASP1
CXCR2
DYSF
CSF3R
S100A9
S100A8
MMP9
FPR1
IFITM2
CXCL8
IL1R2
G0S2
ITGAX
CSF3R
IER3
TLR2
PLAUR
IL1RN
PLAU
UHRF1
CDT1
HIST1H1B
RRM2
HELLS
CENPM
ASF1B
E2F1
H2AFX
HIST1H1C
CD40LG
CCR4
TRAC
CD3E
CD5
SPOCK2
IL32
CD6
LTB
IL2RA
NKG7
KLRC4
GZMB
GNLY
CD8A
PRF1
LAG3
CTSW
SFMBT2
TNFRSF18
KLRK1
NKG7
CCL5
PRF1
GZMA
CST7
KLRD1
ITM2C
APOBEC3G
TRBC2
SPOCK2
TNFRSF25
NPDC1
CD6
LTB
CTLA4
MAF
SATB1
CD38
LTBP3
CPA3
GATA2
MS4A2
SIGLEC6
KIT
SLC18A2
PTGS1
CLU
HPGDS
VWA5A
WFDC2
KRT7
SLPI
SLC34A2
NAPSA
PGC
SFTPA1
LCN2
SFTPC
SFTPB
Clusters
C1 – AM
C2 – AM
C3 – Cycling AM
C4 – Mafb+ MoMac
C5 – Neu
C6 – Neu
C7 – CD8+ T cell
C8 – CD4+ T cell
C9 – CD8+ T cell
C10 – CD8+ T cell
C11 – CD4+ T cell
C12 – Mast cell
C13 – Epithelial cell

```r
markers_neutro <- c("CSF3R","FCGR3B", "NAMPT", "CXCR2", "S100A8", "S100A9")
marker_epi <- c("EPCAM", "KRT8", "CDH1", "MUC1")
Marker_NK <- c("TRDC", "GNLY", "KLRD1", "KLRK1")
markers_AM <- c("C1QA", "C1QB", "C1QC", "FABP4")
markers_mono <- c("CSF1R", "TMEM176B", "NR4A2")
markers_baso <- c("HDC", "MS4A2", "GATA2", "CPA3", "TPSAB1")
markers_CD8 <- c("CD8A", "CLIC3")
Markers_CD4 <- c("CD4", "IL7R")


DotPlot(Cell_10xAnnotate, features = c(markers_AM, markers_mono, "VCAN","MAFB",
markers_neutro, "MKI67", "TOP2A", markers_CD8, Markers_CD4, markers_baso, marker_epi),
group.by = "Annotation") + theme_classic() + scale_size(range = c(0,12)) +
theme(axis.text.x = element_text(angle = 45, hjust = 1), axis.text = element_text(size =
35),
           legend.text   = element_text(size = 25), legend.key.height= unit(1, 'cm'),
        legend.key.width= unit(2, 'cm'))
```

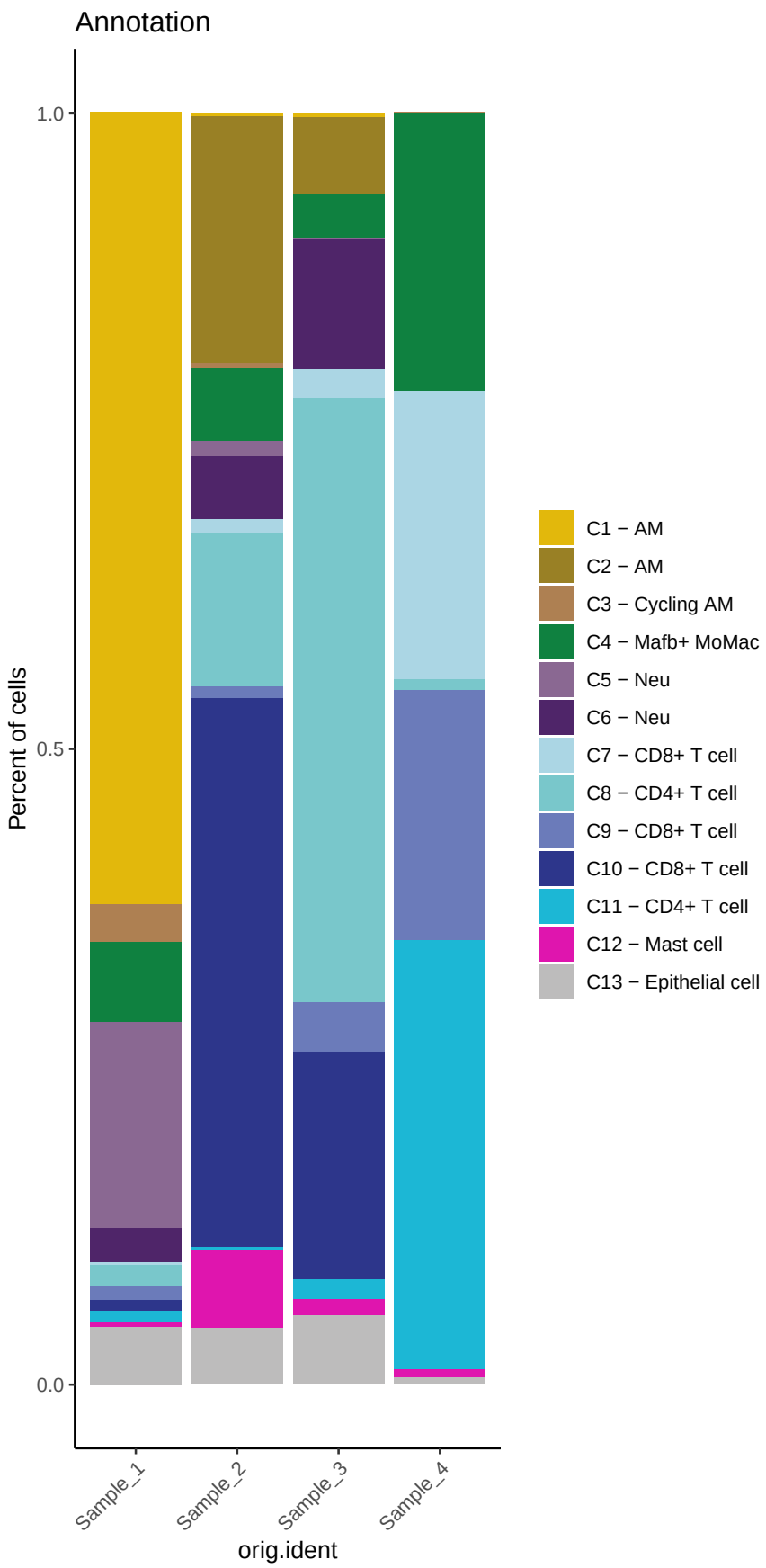
```
## Warning in FetchData.Seurat(object = object, vars = features, cells = cells):
## The following requested variables were not found: TPSAB1
```

```
## Scale for size is already present.
## Adding another scale for size, which will replace the existing scale.
```




```r
#ggsave("Dotplot_annotated.pdf", width = 35, height = 10)
```

```r
var_order <- c(1,6,7,8,9,10,11,12,13,2,3,4,5)
dittoBarPlot(Cell_10xAnnotate, "Annotation", group.by ="orig.ident", var.labels.reorder =
var_order, color.panel = col)
```


Annotation
1.0
0.5
0.0
Percent of cells
Sample_1
Sample_2
Sample_3
Sample_4
orig.ident
C1 − AM
C2 − AM
C3 − Cycling AM
C4 − Mafb+ MoMac
C5 − Neu
C6 − Neu
C7 − CD8+ T cell
C8 − CD4+ T cell
C9 − CD8+ T cell
C10 − CD8+ T cell
C11 − CD4+ T cell
C12 − Mast cell
C13 − Epithelial cell

```r
#ggsave("bar_freq_10x.pdf")
```

```r
#saveRDS(Cell_10xAnnotate, "lavage_10x_annotated.rds")
```

```r
sessionInfo()
```

```
## R version 4.3.3 (2024-02-29)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=fr_BE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=fr_BE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=fr_BE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=fr_BE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Brussels
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
## [1] formatR_1.14          ComplexHeatmap_2.16.0 dittoSeq_1.12.0
## [4] RColorBrewer_1.1-3    ggplot2_3.4.2         patchwork_1.1.2
## [7] SeuratObject_4.1.3    Seurat_4.3.0          dplyr_1.1.2
##
## loaded via a namespace (and not attached):
##   [1] shape_1.4.6                 rstudioapi_0.14
##   [3] jsonlite_1.8.7              magrittr_2.0.3
##   [5] magick_2.7.5                spatstat.utils_3.0-3
##   [7] farver_2.1.1                rmarkdown_2.23
##   [9] GlobalOptions_0.1.2         zlibbioc_1.46.0
##  [11] vctrs_0.6.3                 ROCR_1.0-11
##  [13] Cairo_1.6-2                 spatstat.explore_3.2-1
##  [15] RCurl_1.98-1.12             S4Arrays_1.2.1
##  [17] htmltools_0.5.5             sctransform_0.3.5
##  [19] parallelly_1.36.0           KernSmooth_2.23-22
##  [21] htmlwidgets_1.6.2           ica_1.0-3
##  [23] plyr_1.8.8                  plotly_4.10.2
##  [25] zoo_1.8-12                  igraph_1.5.0.1
##  [27] iterators_1.0.14            mime_0.12
##  [29] lifecycle_1.0.3             pkgconfig_2.0.3
##  [31] Matrix_1.6-1                R6_2.5.1
##  [33] fastmap_1.1.1               clue_0.3-64
##  [35] GenomeInfoDbData_1.2.10     MatrixGenerics_1.12.2
```

```
##  [37] fitdistrplus_1.1-11         future_1.33.0
##  [39] shiny_1.7.4.1               digest_0.6.33
##  [41] colorspace_2.1-0            S4Vectors_0.38.1
##  [43] tensor_1.5                  irlba_2.3.5.1
##  [45] GenomicRanges_1.52.0        labeling_0.4.2
##  [47] progressr_0.13.0            fansi_1.0.4
##  [49] spatstat.sparse_3.0-2       httr_1.4.6
##  [51] polyclip_1.10-4             abind_1.4-5
##  [53] compiler_4.3.3              doParallel_1.0.17
##  [55] withr_2.5.0                 highr_0.10
##  [57] MASS_7.3-60.0.1             DelayedArray_0.26.3
##  [59] rjson_0.2.21                tools_4.3.3
##  [61] lmtest_0.9-40               httpuv_1.6.11
##  [63] future.apply_1.11.0         goftest_1.2-3
##  [65] glue_1.6.2                  nlme_3.1-164
##  [67] promises_1.2.0.1            Rtsne_0.16
##  [69] cluster_2.1.6               reshape2_1.4.4
##  [71] generics_0.1.3              gtable_0.3.3
##  [73] spatstat.data_3.0-1         tidyr_1.3.0
##  [75] data.table_1.14.8           XVector_0.40.0
##  [77] sp_2.0-0                    utf8_1.2.3
##  [79] BiocGenerics_0.46.0         spatstat.geom_3.2-4
##  [81] RcppAnnoy_0.0.21            foreach_1.5.2
##  [83] ggrepel_0.9.3               RANN_2.6.1
##  [85] pillar_1.9.0                stringr_1.5.0
##  [87] limma_3.56.2                spam_2.9-1
##  [89] later_1.3.1                 circlize_0.4.15
##  [91] splines_4.3.3               lattice_0.22-5
##  [93] survival_3.5-8              deldir_1.0-9
##  [95] tidyselect_1.2.0            SingleCellExperiment_1.22.0
##  [97] miniUI_0.1.1.1              pbapply_1.7-2
##  [99] knitr_1.43                  gridExtra_2.3
## [101] IRanges_2.34.0              SummarizedExperiment_1.30.2
## [103] scattermore_1.2             stats4_4.3.3
## [105] xfun_0.39                   Biobase_2.60.0
## [107] matrixStats_1.0.0           pheatmap_1.0.12
## [109] stringi_1.7.12              lazyeval_0.2.2
## [111] yaml_2.3.7                  evaluate_0.21
## [113] codetools_0.2-19            tibble_3.2.1
## [115] cli_3.6.1                   uwot_0.1.16
## [117] xtable_1.8-4                reticulate_1.30
## [119] munsell_0.5.0               Rcpp_1.0.11
## [121] GenomeInfoDb_1.36.0         globals_0.16.2
## [123] spatstat.random_3.1-5       png_0.1-8
## [125] parallel_4.3.3              ellipsis_0.3.2
## [127] dotCall64_1.0-2             bitops_1.0-7
## [129] listenv_0.9.0               viridisLite_0.4.2
## [131] scales_1.2.1                ggridges_0.5.4
## [133] crayon_1.5.2                leiden_0.4.3
## [135] purrr_1.0.1                 GetoptLong_1.0.5
## [137] rlang_1.1.1                 cowplot_1.1.1
```